DOUBAXIL SACHET
GLUCOSAMINE & CHONDROITIN POWDER

Active Ingredients:
Each sachet contains:

Glucosamine Sulphate Sodium Chloride	1884mg
(equivalent to Glucosamine Sulphate 1500mg)	
Derived from seafood	
Chondroitin Sulphate Sodium	1200mg
Derived from chicken	

Description: White or almost white crystalline powder is filled in aluminium sachet. The solution is clear and colourless with salty taste after dissolved the powder in water.

Pharmacodynamic:
Glucosamine is a natural substance found in chitin, mucoproteins, and mucopolysaccharides. It is involved in the manufacture of glycosaminoglycan, which forms cartilage tissue in the body; glucosamine is also present in tendons and ligaments. Glucosamine must be synthesized by the body but the ability to do this decline with ages. Glucosamine and its salts have therefore been advocated in the treatment of rheumatic disorders including osteoarthritis.
Glucosamine also acts to improve the viscosity of synovial fluid by increasing synovial fluid production, thereby providing lubricant activity.

The effect of chondroitin sulfate in patients with osteoarthritis is likely the result of a number of reactions including its anti-inflammatory activity, the stimulation of the synthesis of proteoglycans and hyaluronic acid, and the decrease in catabolic activity of chondrocytes inhibiting the synthesis of proteolytic enzymes, nitric oxide, and other substances that contribute to damage cartilage matrix and cause death of articular chondrocytes. A recent review summarizes data from relevant reports describing the biochemical basis of the effect of chondroitin sulfate on osteoarthritis articular tissues. The rationale behind the use of chondroitin sulfate is based on the belief that osteoarthritis is associated with a local deficiency in some natural substances, including chondroitin sulfate.
Recently, new mechanisms of action have been described for chondroitin sulfate. In an in vitro study, chondroitin sulfate reduced the IL-1β-induced nuclear factor-kB (NF-κB) translocation in chondrocytes. In addition, chondroitin sulfate has recently shown a positive effect on osteoarthritic structural changes occurred in the subchondral bone.

Pharmacokinetic:
Glucosamine
Absorption
After oral administration, bioavailability is low due to first-pass hepatic metabolism 26%. The gastrointestinal absorption is close to 90%.

Distribution
Glucosamine is not protein-bound, but rather incorporates into plasma proteins (primarily globulins). Volume of distribution: 2.5 Liters.

Metabolism
- Liver extensive

The first-pass effect in the liver in which more than 70% of glucosamine is metabolized.

Excretion
Renal Excretion, 10%
Feces, 11%
Part of a dose of glucosamine sulfate is eliminated as carbon dioxide via expired air.

Pharmacokinetic studies performed on humans and experimental animals after oral administration of chondroitin sulfate revealed that it can be absorbed orally.

Chondroitin sulfate shows first-order kinetics up to single doses of 3,000 mg. Multiple doses of 800 mg in patients with osteoarthritis do not alter the kinetics of chondroitin sulfate. The bioavailability of chondroitin sulfate ranges from 15% to 24% of the orally administered dose. More particularly, on the articular tissue reported that chondroitin sulfate is not rapidly absorbed in the gastro-intestinal tract and a high content of labelled chondroitin sulfate is found in the synovial fluid and cartilage.

Indication: Used as adjuvant therapy for osteoarthritis.

Recommended Dose:
Adult: One sachet to be taken once a day. The entire contents of one sachet should be fully dissolved in at least 250 ml of water (one glass) and drink immediately. It is best to be taken before meal for at least 8-12 weeks or more, according to medical prescription.

Route of Administration: Oral

Contraindication:
Hypersensitivity to glucosamine sulphate and chondroitin sulphate.
As the active ingredient is obtained from seafood , the product should not be given to patients who are allergic to seafood.

Warning & Precautions:
This product treats the underlying cause of osteoarthritis and the therapeutic effect can only be seen after a few weeks. Therefore, it is advisable to take an analgesic/anti-inflammatory drug if required during the first few weeks of therapy with this product. Safety and effectiveness have not been established in children (<18 years), therefore children (<18 years) should avoid using this product.

The administration in patients with severe hepatic or renal insufficiency should be made under medical supervision.

A doctor should be consulted in order to exclude the presence of other joint conditions/ diseases for which an alternative treatment should be considered.

Keep out of reach of children.

Effect on Ability to Drive and Use Machines: Not known

Interactions with Other Medicaments:
1) Effects on glucose metabolism & antidiabetic agents: It has been hypothesized that glucosamine may impair insulin secretion through competitive inhibition of glucokinase in pancreatic beta cells and/or alteration of
peripheral glucose uptake. Glucosamine may increase insulin resistance and consequently affect glucose tolerance. It may reduce antidiabetic agent effectiveness e.g. when used with these antidiabetic agents: Acarbose, Acetohexamide, Chlorpropamide, Glipizidede, Glyburide, Metformin, Miglitol, Pioglitazone, Repaglinide, Rosiglitazone, Glimepiride, Tolbutamide, Troglitazone, Glucosamine is likely safe in patients with well-controlled diabetes (HbA1c less than 6.5%) taking one or two oral antidiabetic medications or controlled by diet only. In patients with higher HbA1c levels or those taking insulin, monitor blood glucose levels closely/more frequently.
2) Reduced effectiveness when used with glucosamine: Doxorubicin, Etoposide, Teniposide.
3) Warfarin
- Elevations of International Normalized Ratio serum values and potentiation of anticoagulant effects.
- If concomitant therapy is necessary, the patient's INR should be more closely monitored.
4) Avoid combination with chitosan as it may form complexes with chondroitin sulfate, thus decreasing its absorption.

Pregnancy & Lactation:
Available evidence is inconclusive or inadequate for use in pregnant or lactating mothers. Until more information is available, this product should only be used under medical supervision in pregnancy and lactating mothers if the potential benefit to the mother justifies the potential risk to the fetus.
Administration during the first 3 months of pregnancy must be avoided.

Side Effects:
- Cardiovascular

Peripheral oedema, tachycardia were reported in a few patients following larger clinical trials investigating oral administration in osteoarthritis. Causal relationship has not been established.

- Central nervous system

Drowsiness, headache, insomnia have been observed rarely during therapy (less than 1%).

- Gastrointestinal

Nausea, vomiting, diarrhea, dyspepsia or epigastric pain, constipation, heartburn and anorexia have been described rarely during oral therapy with glucosamine.

- Skin

Skin reactions such as erythema and pruritus have been reported with therapeutic administration of glucosamine.

Symptoms & Treatment of Overdose:
No cases of accidental or intentional overdoses are known. The animal acute and chronic toxicological studies indicate that toxic effects and symptoms of toxicity are not likely to occur, even after high overdoses.

Storage Condition:
This product should be stored below 30°C in a dry place, away from direct sunlight and moisture.

Shelf Life: Please refer to the packaging. Do not use beyond the expiry date.

Packing: 30 sachets x 3084mg per box

Distributed by Product Registration Holder:
Janipro (M) Sdn. Bhd. (838162-X)
42 Jalan Seksyen 1/21, Taman Kajang Utama,
43000 Hulu Langat, Selangor, Malaysia.

Manufacturer:
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81200 Johor Bahru, Johor, Malaysia.

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